

and guides users. The headset needs a smartphone to function and Bluetooth and Wi-Fi to receive information. It is compatible with Windows phone, and the phone app is gesture based, easy-to-use by any visually-impaired person.

Microsoft employee, Jennifer Warrick, tried it out blindfolded and the headset worked perfectly in guiding her using only sounds. The headset uses an orientation and navigation system that helps users stay on course.

An electro-adhesive robotic hand that can make solar cells

Gabit Inc., a start-up, has developed a concept robot that can pick up things using electro-adhesion power. This new concept is based on electrostatic attraction.

The gripper, which is made of flexible materials with electrostatic properties, on Gabit holds things while lifting them. It can adjust its shape according to the object. Gabit is based on the research done by SRI International. The same concept is being used in the industry for handling electronic components, but with gripper technology, it can be used for multiple products and materials.

The gripper technology plays a key role in the Gabit model. The flexibility of the gripper makes it very gentle and perfect in handling delicate materials. The gripper is very energy-efficient, which makes it economical-to-use as compared to electromagnet robot hands.

Gabit Inc. aims to revolutionise the industry with their disruptive technology. The founders claim that Gabit can be used in manufacturing solar cells and micro-PCBs.

A telepathic system that establishes direct connection between brains over the Internet

A scientific study on direct transmission of signals has led to a successful replication of a direct brain-to-brain connection between multiple pairs of people.



Researcher Rajesh Rao, left, plays a computer game with his mind, while researcher Andrea Stocco, right, wears a magnetic stimulation coil over the left motor cortex region of his brain (Courtesy: <http://www.nbcnews.com>)

The six-member research team led by Rajesh Rao, a scientist of Indian origin, has successfully transmitted signals from one brain to control motions of another person, over the Internet using signals. Everything happened within fractions of a second. This is believed to be the closest thing to telepathy. Rao was joined by Andrea Stocco in the research and they did the research in collaboration with UW Institute for Learning & Brain Sciences, Computer Science & Engineering and Department of Psychology.

The team took help from an electroencephalography (EEG) machine and studied Rao's brain when he was gazing at a computer screen and playing a simple video game with just his mind and no hand movement. In a different lab, Stocco was sitting with a transcranial magnetic stimulation coil placed on his left motor cortex to control his hand movement. Whenever Rao imagined that he was hitting the 'fire' button in his game, Stocco automatically moved his right index finger to fire.

Scientists are now exploring opportunities on how brain waves can be influenced in case of alertness and sleepiness. For the last one year, the team has been working on developing the new system and has recently received US\$ 1 million W.M. Keck Foundation grant from University

of Washington.

This technology can be used for 'brain tutoring' in the future for transferring knowledge directly from a teacher's brain to a student.

Here's an app to monitor suicidal plans

A service has been launched by suicide prevention charity Samaritans that can spot people who are likely to be planning to end their own lives. The service has been named Samaritans Radar and can be used by Twitter users as of now.

This service allows monitoring friends' tweets that show signs that they are struggling with their lives. An algorithm is used in this app that detects keywords and phrases like 'hate myself,' 'tired of being alone,' 'depressed,' 'help me,' 'need someone to talk to,' etc. The service alerts a user so that he or she can reach out to the friend in need and offer help.

Twitter's API has been used in developing this app and third-party applications are also allowed to get integrated with this service. The charity organisation plans to get more updates to this app when more data becomes available. This service can make online space a better place for vulnerable people, as told by Joe Ferns, executive director of policy, research and development at Samaritans.

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